Internal SoC Architecture

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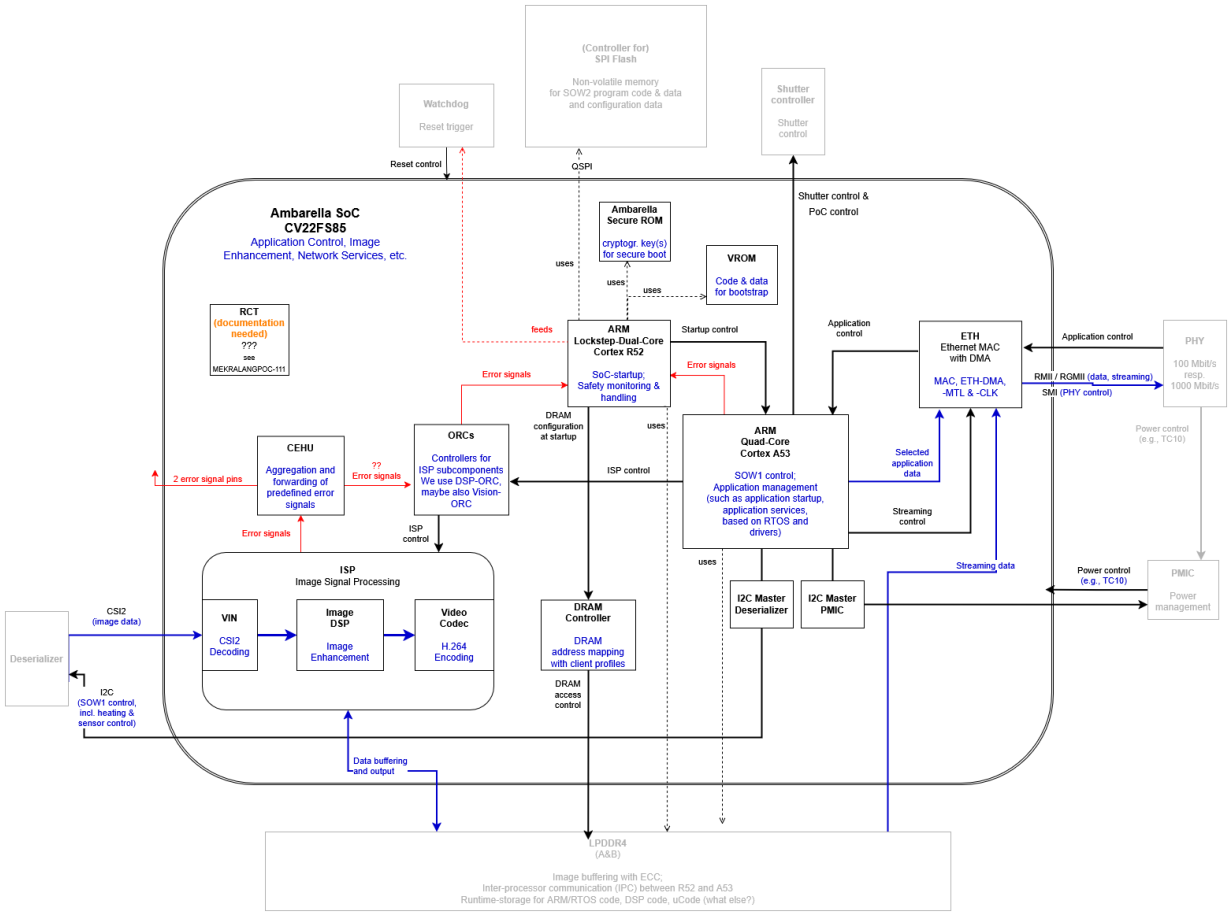
1 SoC Overview

We try to establish an architectural overview of the SoC Hardware components that are related to this project.

The SoC has many different Hardware components, some we cannot directly use, some not well documented, some just not used in this project. So the overview is by no means complete, it is just to establish the important SoC parts that need our attention.

1.1 Major Functional Components

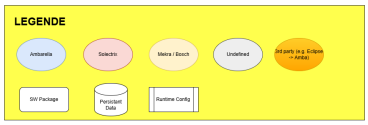
MVC-98290 - SoC major functional components

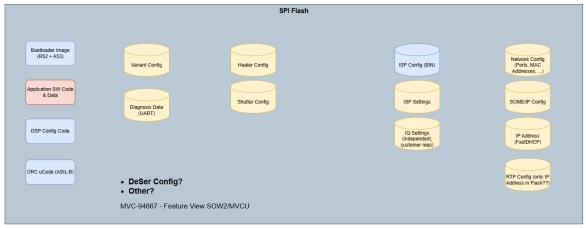


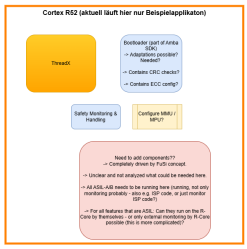
[draft]

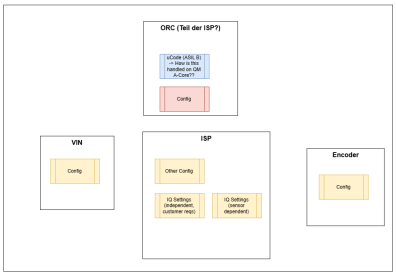
1.2 SW / Data Deployment

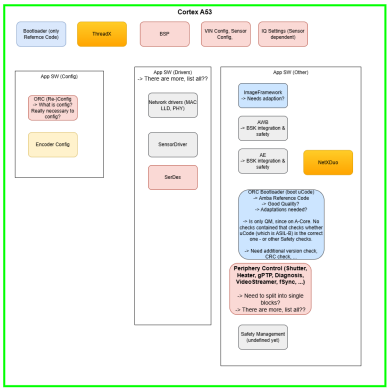
SoC SW / Data Deployment (Draft)

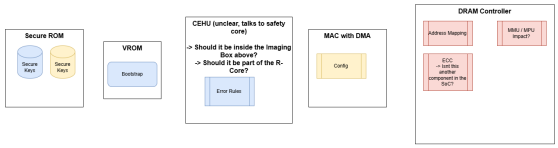












1.3 Boot

1.3.1 ORCs and ISP

ORCs and ISP need Software from Ambarella. This Software must be loaded by the Application Software.

#TODO ein Bild wäre schick. Die Schnittstellen zum laden der uCode Software muss noch identifiziert werden

Application Software muss prüfen, dass die korrekte Software geladen wird. Es ist möglich eine uCode Software zu laden die nicht zum SoC Chip passt. Siehe auch JIRA Ticket:

MEKRALANGPOC-263 Checking ucode version

- Das define ist in dem Header ROOT/rtos/cortex_a/soc/dsp/visual/inc/LL/fs/cv2fs/cmd_msg_dsp.h

- Die Funktion, um die aufgespielte Version abzufragen ist: AmbaDSP_MainGetDspVerInfo

2 SPI-NOR-Controller

The uC SW shall configure the SPI-NOR controller for the following functionality:

1. Disable SM error detection

Offset	Name	Bits	Field	Reset	Val	Descripton
0x5C	SPI-NOR Fault Injection	1	sm_error_disable	0x0	0x1	0x1 = disable SM error (TBD)

2. Set DMA channel to SPI-NOR RX and TX
3. Set SPI-NOR clock divider to match 100 MHz

Offset	Name	Bits	Field	Reset	Val	Descripton
0x08	SPI-NOR Flow Control	17:10	CLKDIVIDER	0x1E	0xXX	to be calculated

4. Disable multiple chip select

Offset	Name	Bits	Field	Reset	Val	Descripton
0x08	SPI-NOR Flow Control	25:18	CHIPSELECT#	0xFE	0x00	disable CEN for multiple devices

5. Enable flow control

Offset	Name	Bits	Field	Reset	Val	Descripton
0x08	SPI-NOR Flow Control	31	FLOWCONTROL	0x1	0x1	0x1 = enable flow control

6. Configure hold clock for flow control

Offset	Name	Bits	Field	Reset	Val	Descripton
0x08	SPI-NOR Flow Control	30:28	HOLD	0x0	0x0	0x0 = gate clock to perform flow control

7. Set RX sample delay

Offset	Name	Bits	Field	Reset	Val	Descripton
0x08	SPI-NOR Flow Control	4:0	RXSAMPLEDELAY	0x0	0x3	0x3 = 3 clock cycle shift

8. Reset FIFO

Offset	Name	Bits	Field	Reset	Val	Descripton
0x40	SPI-NOR TX FIFO Reset	0	TXFIFORESET	0x0	0x1	0x1 = Software reset the TX FIFO
0x44	SPI-NOR RX FIFO Reset	0	RXFIFORESET	0x0	0x1	0x1 = Software reset the RX FIFO

9. Set TX FIFO threshold level

Offset	Name	Bits	Field	Reset	Val	Descripton

Offset	Name	Bits	Field	Reset	Val	Descripton
0x20	SPI-NOR RX FIFO Threshold	8:0	RXFIFOTHLV	0x0	0x1F	0x1F = RX FIFO Threshold level

10. Set RX FIFO threshold level

Offset	Name	Bits	Field	Reset	Val	Descripton
0x1C	SPI-NOR TX FIFO Threshold	8:0	TXFIFOTHLV	0x0	0x1F	0x1F = TX FIFO Threshold level

11. Configure the SPI flash to operate in 4 byte addressing mode by setting EXTADD=1 by issuing Enter 4-byte Address Mode (EN4B = 0xB7)

3 Reading from SPI flash

The uC SW shall read data using the SPI-NOR controller by executing the following sequence:

1. Configure DMA
2. Clear the remain data on receive FIFO if not empt

Offset	Name	Bits	Field	Reset	Val	Descripton
0x40	SPI-NOR TX FIFO Reset	0	TXFIFORESET	0x0	0x1	0x1 = Software reset the TX FIFO
0x44	SPI-NOR RX FIFO Reset	0	RXFIFORESET	0x0	0x1	0x1 = Software reset the RX FIFO

3. Set the SPI-NOR Command Format Restister

Offset	Name	Bits	Field	Reset	Val	Descripton
0x00	SPI-NOR Command Format Restister	31:30	CMDLENGTH	0x1	0x1	command length
		29:27	ADDRLENGTH	0x3	0x4	address length = 4 byte
		26:22	DUMMYLENGTH	0x0	0x7	8 dummy cycles
		21:0	DATALENGTH	0x18	0xX	length to be read

4. Set the SPI-NOR Address Register

Offset	Name	Bits	Field	Reset	Val	Descripton
0x14	SPI-NOR Address Register	31:0	ADDR3:0	0x0	0xX	address to be read

5. Set the SPI-NOR Command Register according to MVC-92857 - Reading

Offset	Name	Bits	Field	Reset	Val	Descripton
0x0C	SPI-NOR Command Register	7:0	CMD0	0x0	0x6B	0x6B = FRQO

6. Set threshold to receive FIFO

Offset	Name	Bits	Field	Reset	Val	Descripton
0x20	SPI-NOR RX FIFO Threshold	8:0	RXFIFOTHLV	0x0	0x1F	0x1F = RX FIFO Threshold level

7. Enable DMA

Offset	Name	Bits	Field	Reset	Val	Descripton
0x18	SPI-NOR DMA Enable	1:0	TXFIFORESET	0x0	0x3	0x3 = enable TX and RX DMA

8. Set transaction done interrupt mask

Offset	Name	Bits	Field	Reset	Val	Descripton
0x30	SPI-NOR Interrupt Mask	5	DATALENTHREACHMSK	0x0	0x1	0x1 = Mask this interrupt

9. Enable transmit/receive bit

Offset	Name	Bits	Field	Reset	Val	Descripton
0x50	SPI-NOR TRX Start	0	STRTTRX	0x0	0x1	0x1 = Start transaction

10. Wait for interrupt transaction done interrupt
11. Read FIFO

Offset	Name	Bits	Field	Reset	Val	Descripton
0x200 - 0x2FF	SPI-NOR RX Data	31:0	RXDATA	-	-	Data

4 Writing to SPI flash

The uC SW shall write data to the external SPI flash using the SPI-NOR controller by executing the following sequence:

1. Sending write enable command according to MVC-92870 - Write enable
2. Configure DMA
3. Clear the remain data on receive FIFO if not empt

Offset	Name	Bits	Field	Reset	Val	Descripton
0x40	SPI-NOR TX FIFO Reset	0	TXFIFORESET	0x0	0x1	0x1 = Software reset the TX FIFO

Offset	Name	Bits	Field	Reset	Val	Descripton
0x44	SPI-NOR RX FIFO Reset	0	RXFIFORESET	0x0	0x1	0x1 = Software reset the RX FIFO

4. Set the SPI-NOR Command Format Restister

Offset	Name	Bits	Field	Reset	Val	Descripton
0x00	SPI-NOR Command Format Restister	31:30	CMDLENGTH	0x1	0x1	command length
		29:27	ADDRLENGTH	0x3	0x4	address length = 4 byte
		26:22	DUMMYLENGTH	0x0	0x0	0 dummy cycles
		21:0	DATALENGTH	0x18	0xX	length to be written

5. Set the SPI-NOR Address Register

Offset	Name	Bits	Field	Reset	Val	Descripton
0x14	SPI-NOR Address Register	31:0	ADDR3:0	0x0	0xX	address to be written

6. Set the SPI-NOR Command Register according to MVC-92773 - Programming

Offset	Name	Bits	Field	Reset	Val	Descripton
0x0C	SPI-NOR Command Register	7:0	CMD0	0x0	0x32	0x32 = PPQ

7. Set threshold to transmit FIFO

Offset	Name	Bits	Field	Reset	Val	Descripton
0x1C	SPI-NOR TX FIFO Threshold	8:0	TXFIFOTHLV	0x0	0x1F	0x1F = TX FIFO Threshold level

8. DMA??

Offset	Name	Bits	Field	Reset	Val	Descripton
0x18	SPI-NOR DMA Enable	1:0	TXFIFORESET	0x0	0x3	0x3 = enable TX and RX DMA

9. Write FIFO

Offset	Name	Bits	Field	Reset	Val	Descripton
0x100 - 0x1FF	SPI-NOR TX Data	31:0	TXDATA	-	-	Data

10. Set transaction done interrupt mask

Offset	Name	Bits	Field	Reset	Val	Descripton
0x30	SPI-NOR Interrupt Mask	5	DATALENTHREACHMSK	0x0	0x1	0x1 = Mask this interrupt

11. Enable transmit/receive bit

Offset	Name	Bits	Field	Reset	Val	Descripton
0x50	SPI-NOR TRX Start	0	STRTTRX	0x0	0x1	0x1 = Start transaction

12. Wait for interrupt transaction done interrupt
13. Check if TXFIFO is empty and if not delay until it is empty

5 Erasing operation on SPI flash

The uC SW shall erase data on the external SPI flash using the SPI-NOR controller by executing the following sequence:

1. Sending write enable command according to MVC-92870 - Write enable
2. Set the SPI-NOR Command Format Restister

Offset	Name	Bits	Field	Reset	Val	Descripton
0x00	SPI-NOR Command Format Restister	31:30	CMDLENGTH	0x1	0x1	command length
		29:27	ADDRLENGTH	0x3	0x4	address length = 4 byte
		26:22	DUMMYLENGTH	0x0	0x0	0 dummy cycles
		21:0	DATALENGTH	0x18	0x0	length

3. Set the SPI-NOR Address Register

Offset	Name	Bits	Field	Reset	Val	Descripton
0x14	SPI-NOR Address Register	31:0	ADDR3:0	0x0	0xX	start address of sector to be erased

4. Set the SPI-NOR Command Register according to MVC-92858 - Erasing

Offset	Name	Bits	Field	Reset	Val	Descripton
0x0C	SPI-NOR Command Register	7:0	CMD0	0x0	0xXX	0x21 = SER, 0xD8 = BER64

5. Set transaction done interrupt mask

Offset	Name	Bits	Field	Reset	Val	Descripton
0x30	SPI-NOR Interrupt Mask	5	DATALENTHREACHMSK	0x0	0x1	0x1 = Mask this interrupt

6. Enable transmit/receive bit

Offset	Name	Bits	Field	Reset	Val	Descripton

Offset	Name	Bits	Field	Reset	Val	Descripton
0x50	SPI-NOR TRX Start	0	STRTTRX	0x0	0x1	0x1 = Start transaction

7. Wait for interrupt transaction done interrupt